Nederlandse norm

# NEN-EN 1005-1+A1

(en)

## Safety of machinery - Human physical performance - Part 1: Terms and definitions

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Vervangt NEN-EN 1005-1:2001

ICS 01.040.13; 13.110-30; 13.180

november 2008

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## Nederlands voorwoord

Voor de in deze norm vermelde normatieve verwijzingen bestaan in Nederland de volgende equivalenten:

| vermelde norm | Nederlandse norm | titel |
|---|---|---|
| EN 292-1:1991 | - | - |
| EN 1005-2 | - | - |
| prEN 1005-3:2001 | - | - |
| EN 1005-4 | - | - |
| EN 1005-5 | NEN-EN 1005-5 | Veiligheid van machines - Menselijke fysieke belasting - Deel 5: Risicobeoordeling voor herhaalde handelingen met hoge frequentie (en) |

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EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

# EN 1005-1:2001+A1

October 2008

ICS 01.040.13; 13.110; 13.180

Supersedes EN 1005-1:2001

English Version

# Safety of machinery - Human physical performance - Part 1: Terms and definitions

Sécurité des machines - Performance physique humaine - Partie 1: Termes et définitions

Sicherheit von Maschinen - Menschliche körperliche Leistung - Teil 1: Begriffe

This European Standard was approved by CEN on 30 September 2001 and includes Amendment 1 approved by CEN on 18 August 2008.

CEN members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration. Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the CEN Management Centre or to any CEN member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CEN member into its own language and notified to the CEN Management Centre has the same status as the official versions.

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# Foreword

This document (EN 1005-1:2001+A1:2008) has been prepared by Technical Committee CEN/TC 122 "Ergonomics", the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by April 2009, and conflicting national standards shall be withdrawn at the latest by December 2009.

This document includes Amendment 1, approved by CEN on 2008-08-18.

This document supersedes EN 1005-1:2001.

The start and finish of text introduced or altered by amendment is indicated in the text by tags ▌A1⟩ ⟨A1▌.

This document has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

▌A1⟩ For relationship with EU Directive(s), see informative Annexes ZA and ZB, which are integral parts of this document. ⟨A1▌

EN 1005 consists of the following Parts, under the general title "Safety of machinery – Human physical performance:

— Part 1: Terms and definitions;

— Part 2[1)]: Manual handling of machinery and component parts of machinery;

— Part 3: Recommended force limits for machinery operation;

— Part 4[1)]: Evaluation of working postures and movements in relation to machinery;

— Part 5[1)]: Risk assessment for repetitive handling at high frequency.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

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[1)] This European Standard is under preparation by CEN/TC 122/WG 4 "Biomechanics".



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# Introduction

Ergonomically designed work systems enhance safety, effectiveness and efficiency, improve human working and living conditions, and counteract adverse effects on human health and performance. Good ergonomic design therefore exerts a favourable influence on the work system, and on the reliability of the human being within it.

This European Standard is one of several ergonomic standards for the safety of machinery.

EN 614-1 describes the principles manufacturers should adopt in order to take account of ergonomic factors. This standard has been prepared to be a harmonised standard in the sense of the Machinery Directive and associated EFTA regulations.

This European Standard is a type B standard as stated in EN 1070. The provisions of this document may be supplemented or modified by a type C standard.

NOTE For machines which are covered by the scope of a type C standard and which have been designed and built according to the provisions of that standard, the provisions of that type C standard take precedence over the provisions of this type B standard.

This European Standard has five parts that are intended to cover the range of human physical performance variables relating to machinery design. The scope contained within each document defines the boundaries of their application. Figure 1 illustrates the relationship between the standards concerning the different aspects of human performance. The relationship between standards proceed from those between the different aspects of human physical performance: body dimensions directly influence the shape of postures and movements as well as the available muscle strength, which further vary with postures and movements.

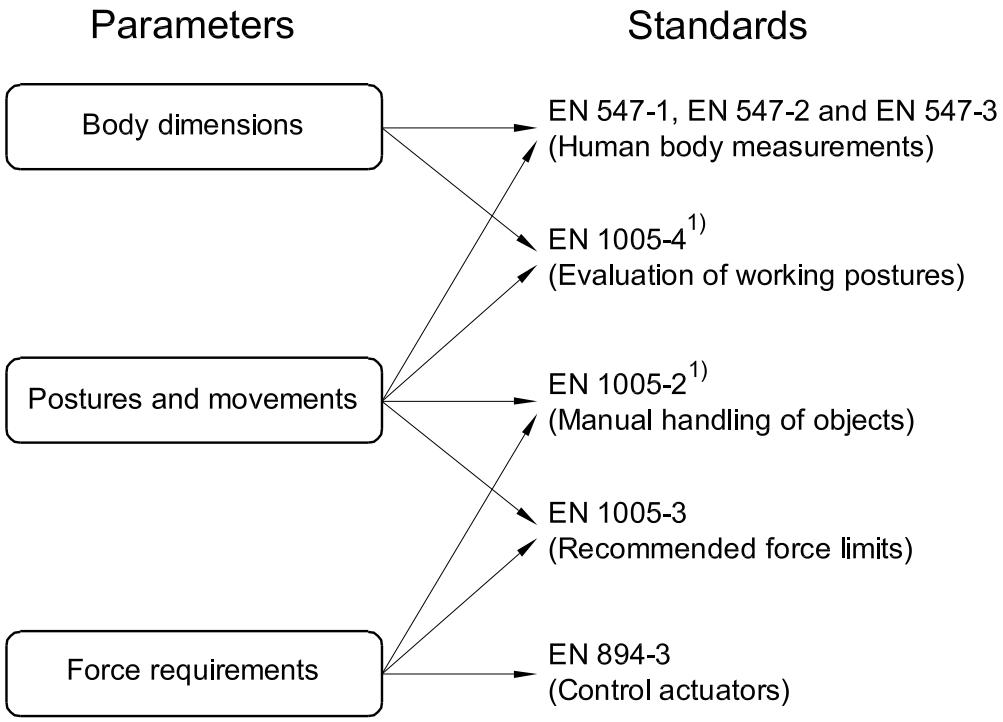


**Figure 1 — Relation between physical performance parameters and standards**

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EN 1005-2[1)], EN 1005-3 and EN 1005-4[1)] adopt a risk assessment approach. The procedures in each part of the standard are different.

The study of human physical loading whilst handling/operating machinery relies on methods and techniques developed by different research disciplines (e.g. epidemiology, biomechanics, physiology, psychophysics). This document aims to define relevant concepts to further co-operation between research disciplines in this field, and to improve the application of standards when designing machinery.

## 1 Scope

This European Standard provides terms and definitions on concepts and parameters used for EN 1005-2[1)], prEN 1005-3, EN 1005-4[1)] and EN 1005-5[1)].

Basic concepts and general ergonomic principles for the design of machinery are dealt with in EN 292-1, EN 292-2 and EN 614-1.

This document is not applicable to specify the machinery which is manufactured before the date of publication of this document by CEN.

## 2 Normative references

This European Standard incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies (including amendments).

EN 292-1:1991, *Safety of machinery - Basic concepts, general principles for design - Part 1: Basic terminology, methodology.*

EN 1005-2[1)], *Safety of machinery – Human physical performance – Part 2: Manual handling of machinery and component parts of machinery.*

prEN 1005-3:2001, *Safety of machinery – Human physical performance – Part 3: Recommended force limits for machinery operation.*

EN 1005-4[1)], *Safety of machinery – Human physical performance – Part 4: Evaluation of working postures and movements in relation to machinery.*

EN 1005-5[1)], *Safety of machinery – Human physical performance – Part 5: Risk assessment for repetitive handling at high frequency.*

## 3 Terms and definitions

For the purposes of this European Standard, the terms and definitions given in EN 1070 together with the following apply.

**3.1**
**abduction**
movement of a limb away from the median plane. See Figure 2 and Figure 3

**3.2**
**action**
the activation of a muscle(s) during a task in order to perform a task/operation (as opposed to rest)

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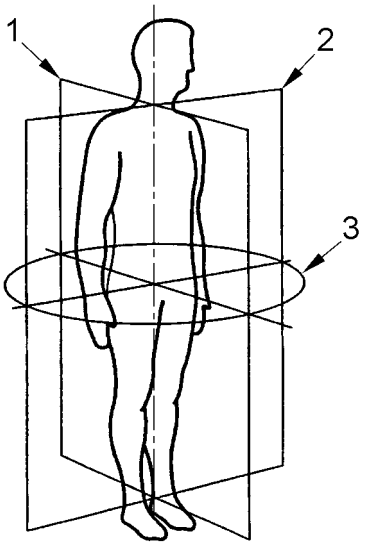
**3.3**
**action period**
the duration of a single cycle of a recurrent event which includes both the rest and action time

**3.4**
**adduction**
movement of a limb towards the median plane. See Figure 2 and Figure 3



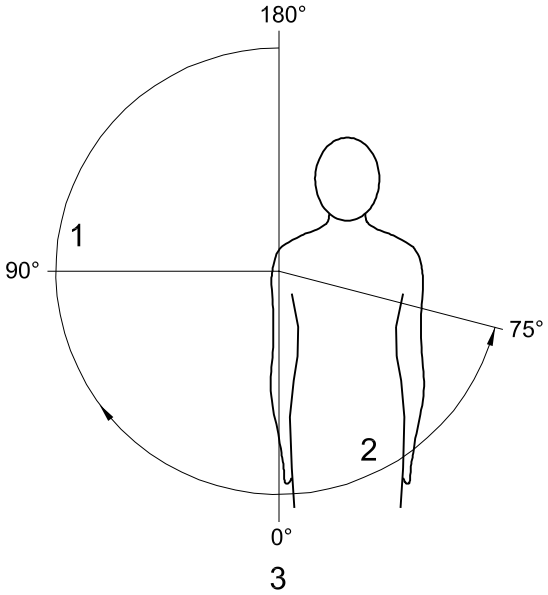

**Key**

1 median or sagittal plane

2 coronal or frontal plane

3 transverse or horizontal plane

**Figure 2 — Planes**




**Key**

1 abduction

2 adduction

3 neutral

**Figure 3 — Abduction and adduction**

**3.5**
**angle of asymmetry**
the angle formed between the lines that result from the intersections of the mid-sagittal plane and the plane of asymmetry. If the feet are repositioned during the lift/lower sequence, the referent planes shall be determined at the point in the action sequence wherein the largest degree of asymmetrical twist is encountered. See Figure 4



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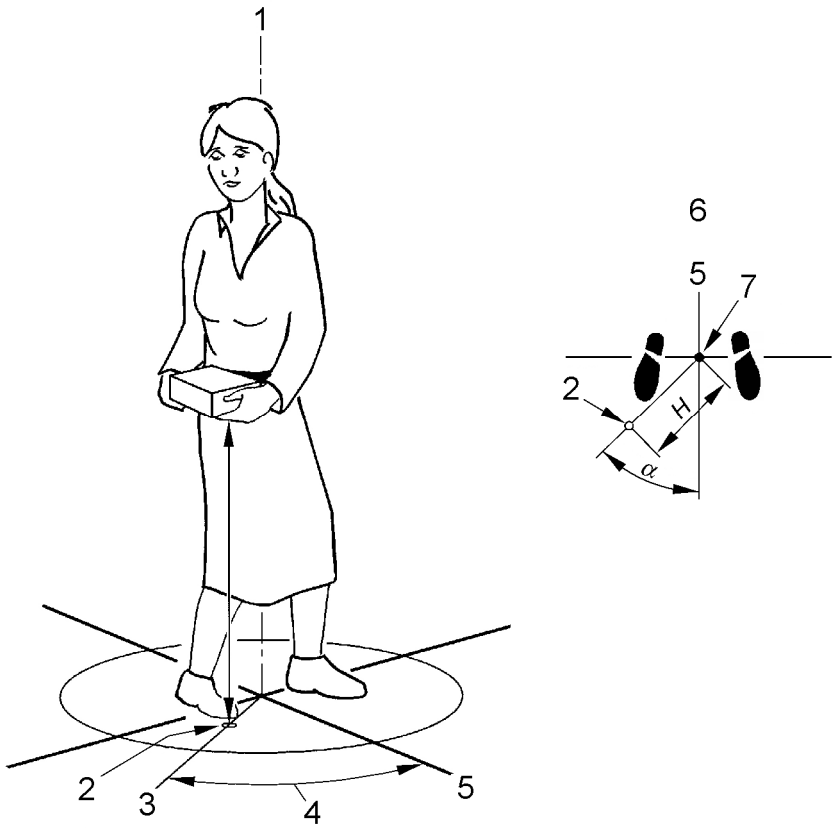


**Key**

1 vertical

2 projection from centre of gravity of load

3 asymmetry line

4 asymmetry angle ($\alpha$)

5 mid-sagittal plane

6 top view

7 mid-point between inner ankle bones

**Figure 4 — Angle of asymmetry**

**3.6**
**general working population**
the adult working population which does not include:
the physically disadvantaged and those under the legal minimum working age

**3.7**
**grip of object**
the manner by which objects can be handled (held and/or moved with the hands). The type of grip applied (e.g. pinch grip, hook grip, power grip), and the design and placement in relation to the characteristics of both the task and the object handled will determine the degree of difficulty of the handling task



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**3.7.1**
**pinch grip**
grip made with the thumb and the forefinger. See Figure 5

**3.7.2**
**hook grip**
with the hook grip the fingers are active and the thumb has a passive role. The proximal and distal interphalangeal joints are flexed around the handle. See Figure 5

**3.7.3**
**power grip**
grip in which fingers and thumb are opposed and clasped around the object to provide maximal surface contact with the palm. This grip serves mainly to apply high forces or to prevent objects from rotating. See Figure 5

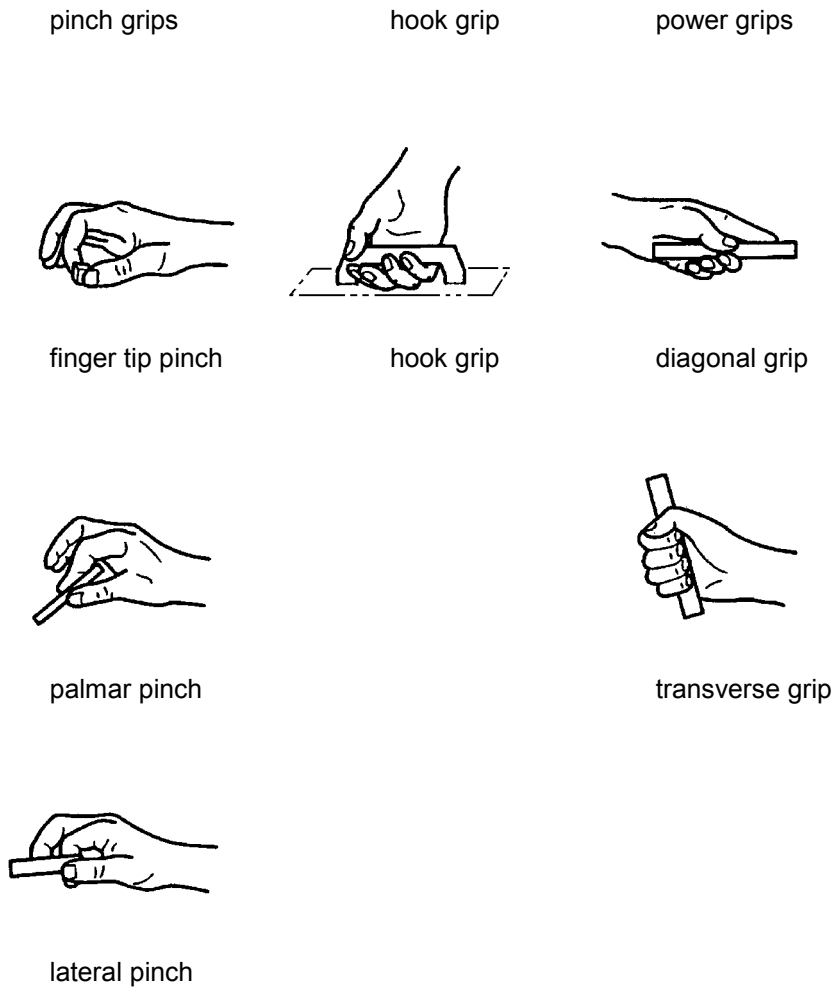


**Figure 5 — Types of grip**

**3.8**
**manual handling**
any activity requiring the use of human force to lift, lower, carry or otherwise move or restrain any object



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**3.8.1**
**manual lifting**
an object is said to be lifted by human force when it is moved from its initial position upwards or downwards

**3.8.2**
**manual carrying**
when an object remains lifted and is moved horizontally by human force, it is carried

**3.8.3**
**horizontal location (*H*)**
horizontal distance of the midpoints of the hands away from the mid-point between the ankles, measured at the origin and destination of lift. See Figure 6

**3.8.4**
**vertical location (*V*)**
vertical distance of the midpoints of the hands above the floor, measured at the origin and destination of a lift. See Figure 6

**3.8.5**
**vertical displacement *(D*)**
absolute value of the difference between the vertical heights at the destination and the origin of the lift. See Figure 6

**3.8.6**
**reference mass**
recommended mass (kg) to be used in the risk assessment method related to an intended user population

**3.8.7**
**actual mass**
mass of the object(s) to be handled manually (kg)

**3.9**
**operator**
the person or persons given the task of installing, operating, adjusting, maintaining, cleaning, repairing or transporting machinery (see 3.21 of EN 292-1:1991). For the purpose of this European Standard the assembling and the dismantling of the machinery is also included in the task of the operator

**3.10**
**posture**
the position of the body, a/the body segment(s) or joint(s)

**3.11**
**recovery time**
the period of rest following a period of activity in which restoration of a muscle(s) can occur



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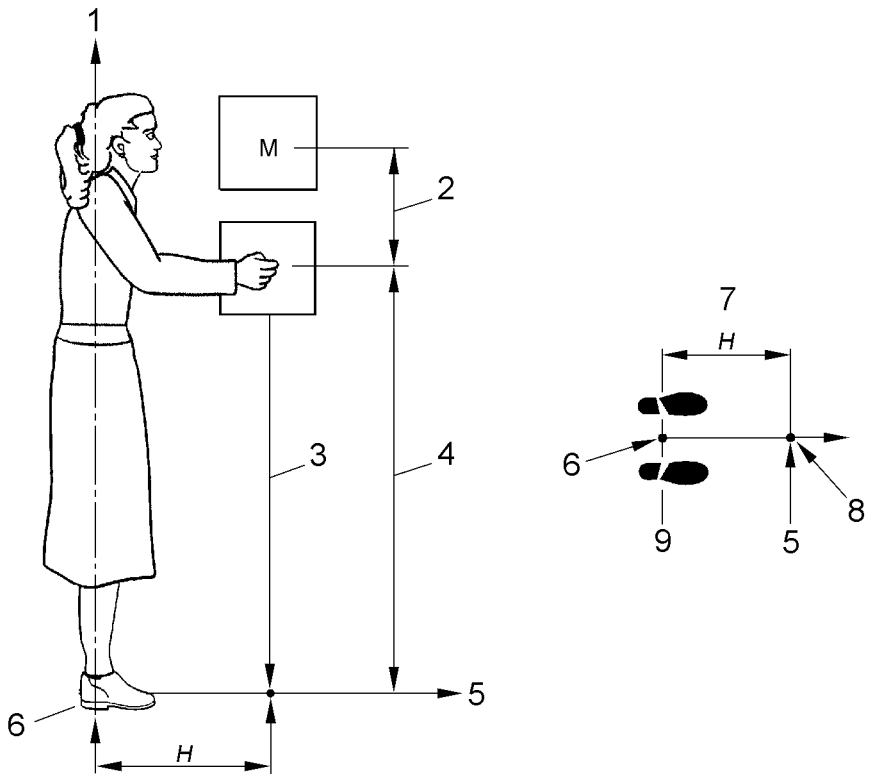


**Key**

1 vertical

2 vertical travel displacement

3 projection from centre of gravity of load

4 vertical location

5 horizontal

6 mid-point between inner ankle bones

7 top view

8 centre of load

9 lateral

**Figure 6 — Horizontal and vertical location**

**3.12**
**rest**
when during a task, no specific task-related activation of muscle is necessary

**3.13**
**risk assessment**
a comprehensive estimation of the probability and the degree of the possible injury or damage to health in a hazardous situation in order to select appropriate safety measures (see 3.8 of EN 292-1:1991 and EN 1050)

**3.14**
**sagittal**
the anteroposterior (front to back) median plane of the body (mid-sagittal), or to a plane parallel to the median (see Figure 2)



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**3.15**
**shift**
the specified minimum number of hours to be worked by an individual in one working day to meet contractual requirements. The number of hours varies between different industries, but is normally between four hours and eight hours in each 24 hours period

**3.16**
**static posture**
a posture maintained for longer than four seconds; this applies to slight or none variations around a fixed force level delivered by muscles and other body structures

**3.17**
**technical aids**
devices for the construction, transport and commissioning, use and dismantling of the machine (e.g. handles, wedges, wheels, ledges, levers, conveyors, cranes, trucks, lifting tables, etc.) which totally or partially eliminate the need for lifting or heavy manual handling, or improve the handling condition, and hence reduce the strain on the body



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# Annex ZA
(informative)

## A1 Relationship between this European Standard and the Essential Requirements of EU Directive 98/37/EC, amended by 98/79/EC

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association to provide a means of conforming to Essential Requirements of the New Approach Directive 98/37/EC on machinery, amended by 98/79/EC.

Once this standard is cited in the Official Journal of the European Communities under that Directive and has been implemented as a national standard in at least one Member State, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the relevant Essential Requirements of that Directive and associated EFTA regulations.

**Table ZA.1 — Correspondence between this European Standard and Directive 98/37/EC, amended by 98/79/EC**

| Clause(s)/sub-clause(s) of this EN | Essential Requirements (ERs) of Directive 98/37/EC, amended by 98/79/EC | Qualifying remarks/Notes |
|---|---|---|
| All clauses | Annex I: 1.1.2 d), 1.1.5 | - |

**WARNING** — Other requirements and other EU Directives may be applicable to the product(s) falling within the scope of this standard. A1



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# Annex ZB
(informative)

## A1 Relationship between this European Standard and the Essential Requirements of EU Directive 2006/42/EC

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association to provide a means of conforming to Essential Requirements of the New Approach Directive 2006/42/EC on machinery.

Once this standard is cited in the Official Journal of the European Communities under that Directive and has been implemented as a national standard in at least one Member State, compliance with the normative clauses of this standard given in Table ZB.1 confers, within the limits of the scope of this standard, a presumption of conformity with the relevant Essential Requirements of that Directive and associated EFTA regulations.

**Table ZB.1 — Correspondence between this European Standard and Directive 2006/42/EC**

| Clause(s)/sub-clause(s) of this EN | Essential Requirements (ERs) of Directive 2006/42/EC | Qualifying remarks/Notes |
|---|---|---|
| All clauses | Annex I: 1.1.5, 1.1.6 | - |

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[4] EN 547-2, *Safety of machinery – Human body measurements – Part 2: Principles for determining the dimensions required for access openings.*

[5] EN 547-3, *Safety of machinery – Human body measurements – Part 3: Anthropometric data.*

[6] EN 614-1, *Safety of machinery – Ergonomic design principles - Part 1: Terminology and general principles.*

[7] EN 894-3, *Safety of machinery – Ergonomics requirements for the design of displays and control actuators – Part 3: Control actuators.*

[8] EN 1050, *Safety of machinery – Principles for risk assessment.*

[9] EN ISO 7250, *Basic list of definitions of human body measurements for technical design (ISO 7250:1996).*

[10] EN 1070, *Safety of machinery – Terminology.*

[11] ENV 26385, *Ergonomic principles of the design of work systems (ISO 6385:1981).*

[12] 90/269/EEC. Council Directive of 29 May 1990 on the minimum health and safety requirements for the manual handling of loads where there is a risk particularly of back injury to workers (fourth individual Directive within the meaning of Article 16 (1) of Directive 89/391/EEC)

[13] 89/391/EEC, Council Directive of 12 June 1989 on the introduction of measures to encourage improvements in the safety and health of workers at work

[14] 89/656/EEC, Council Directive of 30 November 1989 on the minimum health and safety requirements for the use by workers of personal protective equipment at the workplace (third individual Directive within the meaning of Article 16 (1) of Directive 89/391/EEC)

[15] 89/686/EEC, Council Directive of 21 December 1989 on the approximation of the laws of the Member States relating to personal protective equipment (amended by Council Directive 93/95/EEC)

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